FROM TO MERGER TO METAL? TIMESCALES OF R-PROCESS ENRICHMENT FROM NEUTRON STAR MERGERS

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Abstract

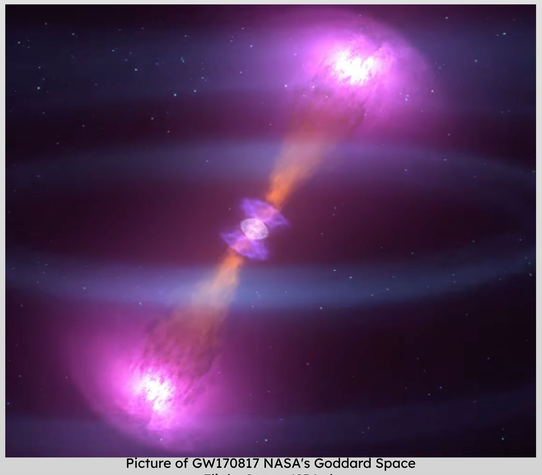
Neutron star mergers (NSMs) are the only confirmed astrophysical sites of r-process nucleosynthesis, yet it remains unclear whether they merge and enrich star-forming gas quickly enough to explain the heavy-element abundances observed in the Milky Way. To test this, I modeled r-process injection and mixing timescales and compared them to observational constraints from short gamma-ray bursts (GRBs). I assembled a sample of 75 short GRBs, including 12 with confirmed kilonovae, and derived enrichment timescales for 87 total events. After optimizing the sample selection, I found that GRBs within 4 kpc of their host-galaxy centers are far more likely to enrich star-forming regions, yielding a median enrichment timescale of approximately 218 Myr, compared to ~1200 Myr for the full sample. Because prior work suggests enrichment requires at least 200 Myr, these results support the idea that NSMs can deliver r-process material to star-forming gas on timescales short enough to account for the observed Galactic heavy-element inventory.

Debate?

The issue is NSMs are delayed events, taking > 100 Myr just to merge. They also can be very offset, typically happening in regions of minimal star formations. Previous research done in the field (Reticulum II by Alexander Ji) gives us reason to believe that r-process elements can be transported and enriched in stars at shorter timescales—a few Myr at most. Since we have very little evidence of NSMs/GRBs occurring in low mass galaxies. A few questions begin to arise: What is the enrichment timescale of r-process from NSMs to star forming gas? Can we computationally prove that NSMs are the dominant source r-process in our universe?

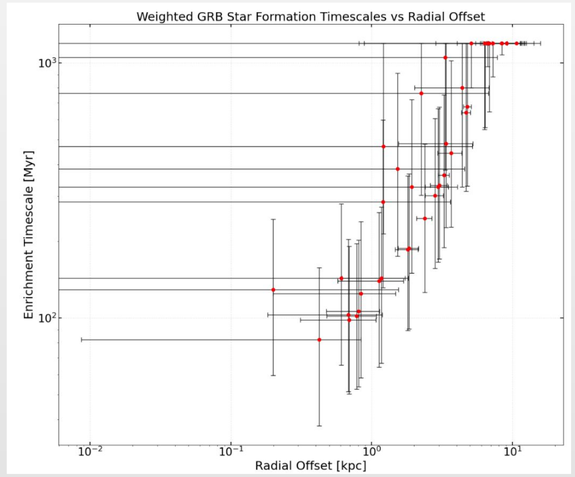
Introduction

The rapid neutron capture process (r-process) occurs when atomic nuclei capture neutrons faster than they can undergo beta decay. This process requires extremely energetic environments with an intense flux of free neutrons, such as neutro nstar mergers or certain rare types of supernovae. Neutron-star mergers (NSMs) are currently the only astrophysical sites where r-process nucleosynthesis has been directly observed. These events occur when two neutron stars lose angular momentum through gravitational-wave emission and eventually collide. The merger produces a short gamma-ray burst (GRB) and is followed by a kilonova (KN), powered by the radioactive decay of freshly synthesized r-process elements. These heavy elements are then ejected into the surrounding interstellar and circumgalactic medium.

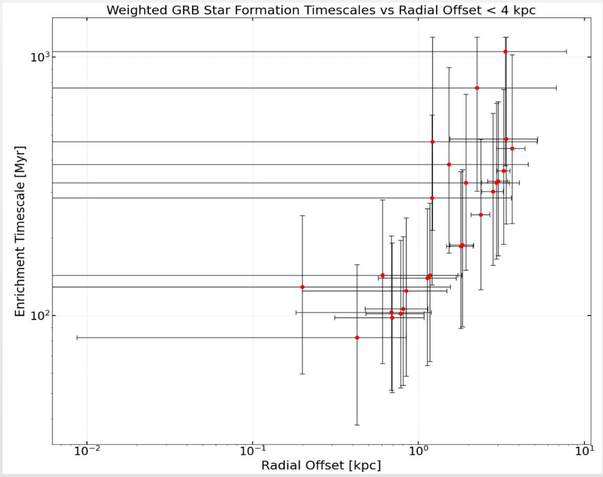


Picture of GW170817 NASA's Goddard Space Flight Center/CI Lab

Results



Results Figure 1: This plot shows the our entire sample of GRB events, with enrichment timescales plotted against their galactocentric offsets. This visualization highlights how positional uncertainty in the GRB offsets propagates into uncertainty in the resulting enrichment timescales. GRBs with larger offsets tend to correspond to longer enrichment timescales. As a result, the median timescale for the entire sample is skewed upward to approximately 1200 Myr. This value corroborates the idea most neutron star mergers occur far from active star-forming regions, where mixing into dense gas proceeds slowly. However, this global median does not capture the full picture.



Results Figure 2: This plot focuses on GRB events occurring at smaller galactocentric offsets, where star formation activity is typically higher. In these regions, r-process material has a significantly greater probability of reaching star-forming gas on shorter timescales. For this subset, the median enrichment timescale drops to approximately 218 Myr. These results support the idea that NSMs can enrich their environments rapidly enough to explain early r-process signatures. Provided the mergers occur sufficiently close to the central, star-forming regions of the galaxy.

Methodology

We used Lagrangian passive flow tracer particles in a galactic simulation to determine how long it takes gas parcels to reach star-forming conditions. By combining each tracer's initial position with the galactocentric offsets of our short gamma-ray burst (GRB) sample (75 short GRBs, including 12 with associated kilonovae), we estimated the enrichment timescales between an NSM event and the incorporation of r-process material into star-forming gas.

Each GRB position contains an intrinsic localization uncertainty, which must be accounted for when estimating the true site of r-process nucleosynthesis. Because kilonova ejecta typically propagate up to ~300 parsecs from the merger site, we adopt 300 pc as the maximum positional uncertainty. For each GRB, we construct a spherical shell bounded by this uncertainty range and use it to approximate the region in which the r-process material was formed and deposited. These shells are then convolved with tracer trajectories to determine the distribution of enrichment timescales.

Conclusion

Our analysis shows that enrichment timescales from neutron-star mergers depend strongly on their galactocentric position. When considering the our full sample of short GRB events, the median enrichment timescale is long (~1200 Myr), reflecting that most NSMs occur at large offsets where mixing into star-forming gas is inefficient. However, restricting the sample to low offset events, those occurring closer to the central, high–star-formation regions reveals substantially shorter enrichment timescales. With a median of ~218 Myr, these results demonstrate that NSMs are capable of enriching their environments on timescales compatible with early r-process signatures, but only when the mergers occur in regions where gas is dense and actively forming stars. Thus, the spatial distribution of NSMs within a galaxy plays a critical role in determining whether their r-process ejecta can contribute to rapid chemical enrichment..

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